

Mubeen Afzal

Graduate Student @ Saarland University, Saarbrücken Germany

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Summary

AI Engineer and Master's student with a focus on bridging the gap between scalable ML systems and secure automation. Hands-on experience developing and deploying high-performance models for Computer Vision and NLP, optimized for threat detection and real-world system resilience. Now seeking to apply this blend of security and AI expertise to the future projects.

Skills

- **GenAI & NLP:** LLM Finetuning (BERT, Gemma, LLaMA), RAG, Vector DBs (Pinecone, Qdrant), Agentic AI, Semantic Search, Text Classification.
- **Programming:** Python (Flask, FastAPI), JavaScript, C, Assembly (Reverse Engineering)
- **AI & ML Frameworks:** PyTorch, TensorFlow, Transformers, Diffusers, LangChain, LangGraph, MLFlow, NLTK
- **Cloud & DevOps:** AWS (SageMaker, Lambda, EC2), Docker, Git, CI/CD pipelines
- **Security:** Anomaly Detection, Threat Intelligence, Vulnerability Assessment
- **Databases:** Elasticsearch, MySQL, Firebase, MongoDB
- **Additional Skills:** English (IELTS: C1), Adept at conveying complex technical concepts to stakeholders, Experienced in cross-functional collaboration

Work Experience

1. Research Intern | CISA Helmholtz Center for Information Security, Saarbrücken

Nov 2025 – Feb 2026

- Developing an automated vulnerability assessment framework that validates web application configurations and orchestrates dynamic crawlers to detect security flaws.
- Build an Agentic AI Framework to find the issues and revive the docker application.

2. AI Engineer | Ikonic, Islamabad

Jun 2025 – Aug 2025

- **Multi-Agent Systems:** Architected autonomous agent workflows using **LangChain**, **LangGraph**, and **Model Context Protocol (MCP)**, enabling complex agent-to-agent communication and task delegation.
- **Process Automation:** Developed scalable low-code/no-code (**N8N**) automation pipelines to streamline repetitive enterprise tasks, significantly reducing operational overhead.
- **System Interoperability:** Engineered modular integration standards to ensure seamless communication between disparate AI agents and external tools.

3. AI Engineer | Imarat Group of Companies, Islamabad

May 2024 – Nov 2024

- **Generative AI for Design:** Pioneered an AI-powered architectural tool utilizing **Diffusion Models** and **ControlNet**, enabling the automated generation of interactive floor plans and photorealistic style rendering.
- **Spatial Layout Optimization:** Leveraged **Graph Neural Networks (GNNs)** to generate and optimize spatial layouts, incorporating style-based customization features for client-specific architectural visualization.
- **Research-to-Product:** Transformed experimental generative models into a functional tool for the real estate design process (Basis for current publication in preparation).

4. Deep Learning Engineer | Qlu.ai, Islamabad

Jan 2024 – Apr 2024

- **Codebase Optimization:** Refactored legacy AI service architectures to improve scalability, reducing latency and simplifying integration with external API ecosystems.
- **Production RAG Systems:** Designed and deployed a **Retrieval-Augmented Generation (RAG)** chat system using vector databases for efficient data retrieval and transformer models for context-aware response generation.
- **MLOps & Monitoring:** Implemented robust model monitoring pipelines to track performance drift and reliability in production environments.

5. Computer Vision Research Intern | DataInsight Lab, Islamabad

Jun 2023 – Aug 2023

- **Medical OCR Pipeline:** Built high-precision multilingual handwritten text recognition pipelines for digitizing medical records, utilizing custom-tuned **YOLO** architectures.
- **Model Optimization:** Enhanced detection accuracy through advanced preprocessing techniques, synthetic data augmentation, and multiple validation strategies to handle noisy real-world data.

Education

M.Sc. Cybersecurity | Saarland University, Saarland | 2025 - Present

BS Artificial Intelligence | NUCES, Islamabad | 2019 – 2024

Certifications

- **Generative AI with LLMs** | DeepLearning.ai
- **Google Cybersecurity Professional Certificate** | Google
- **Azure Fundamentals (AZ-900)** | Microsoft
- **Fundamentals of Privacy Acts and Laws** | Infosec

Publications

- **Afzal, M., et al. (2025). AI-Enhanced Architectural Design: From Interactive Floor Plan Generation to Style Rendering Using Diffusion Models and ControlNet** [Manuscript in preparation]